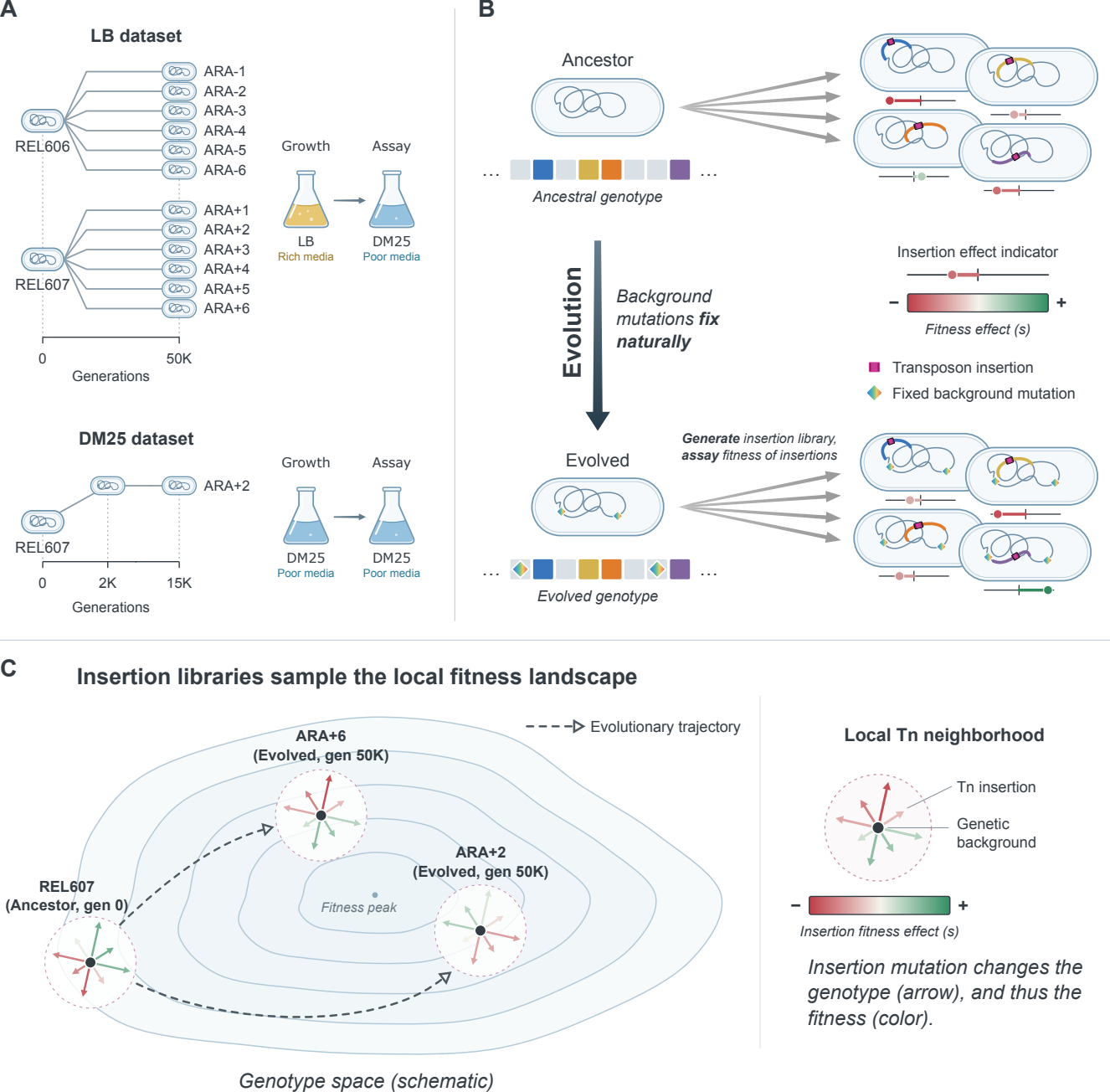

A
LB dataset
REL606
ARA-1
ARA-2
ARA-3
ARA-4
ARA-5
ARA-6
REL607
ARA+1
ARA+2
ARA+3
ARA+4
ARA+5
ARA+6
0
50K
Generations
Growth
Assay
LB
Rich media
DM25
Poor media
DM25 dataset
REL607
ARA+2
0
2K
15K
Generations
Growth
Assay
DM25
Poor media
DM25
Poor media
B
Ancestor
Ancestral genotype
Evolution
Background mutations fix naturally
Evolved
Evolved genotype
Generate insertion library, assay fitness of insertions
Insertion effect indicator
Fitness effect (s)
Transposon insertion
Fixed background mutation
C
Insertion libraries sample the local fitness landscape
Evolutionary trajectory
ARA+6
(Evolved, gen 50K)
ARA+2
(Evolved, gen 50K)
REL607
(Ancestor, gen 0)
Fitness peak
Genotype space (schematic)
Local Tn neighborhood
Tn insertion
Genetic background
Insertion fitness effect (s)
Insertion mutation changes the genotype (arrow), and thus the fitness (color).